Gold Price Prediction Using Time Series Forecasting and Machine Learning Techniques

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1. OVERVIEW

Forecast of gold prices is of high importance in the domain of finance and investments. Beneficiaries like investors, traders, and financial institutions usually depend on price forecasts to take informed decisions about their investment plans as well as risk management issues. Especially in times of global instability, economic uncertainty and inflation rate, gold is seen as one of the safe investment vehicles. Hence, investigating the trends in gold prices and analyzing its future course will be useful for a real-world financial planning.

The main purpose of this project is to predict gold prices based on the historical data and machine learning methodologies. The dataset is collected from Yahoo Finance which has daily data about the price of gold. Project starts with data pre-processing that comprises of missing values handling, testing for stationarity and preparing data for further analysis. Techniques like lag values, rolling statistics and returns are employed for feature engineering to model important price patterns.

Following preprocessing, a machine learning regression model is built on historical gold price data in order to forecast future prices. The performance of the model is calculated using error metrics like MAE, RMSE and R^2 score. Furthermore, F1 score is used to assess the model ability determine the price movement direction. This project shows how machine learning and time series forecasting could be deployed on financial data for worthwhile forecast generation.

2. INTRODUCTION

Financial markets demonstrate complex and evolving patterns that are subject to various economic, political, and social factors. When it comes to types of investment, over time, gold has always emerged as one of those important instruments; this is mainly because of its stability, intrinsic position and inflation-hedging capacity. Many traders use the trend in gold price history often find it easier to predict/anticipate/infer future directions in prices and hence they manage risks much better when dealing with financial instruments.

Sequential financial data including stock prices, commodity prices as well as economic indicators are best studied through time series analysis. Time series data is analyzed by machine learning techniques since they can capture non-linear relationships and patterns. Nevertheless, incomplete values, noise, irregular time intervals and non-stationary behavior which are frequently present in raw financial datasets may degrade the performance of machine learning models.

As the quality and structure of dataset are critical, this paper attempts to review predictive modelling system for the business applications. ML models cannot be trained properly if the data is not clean, well-structured, and normalized. Thus, data preprocessing and feature engineering provide a basis for knowl-edge discovery process – conversion of raw financial data into features that capture statistic rela-tionships and/or temporal patterns.

This project aims to do time series data preprocessing on the gold price data that comes from Yahoo Finance API by using Python. The dataset consist of daily gold price values and is to be analyzed for patterns and trends using statistical as well as visualization techniques. Stationary testing using Augmented Dickey-Fuller test is performed to see whether the dataset satisfies the assumptions required for time-series modelling or not. Difference techniques are then applied to remove trends and stabilize the dataset.

Additionally, feature engineering techniques are used to extract meaningful information from the dataset. Lag features are generated to capture historical dependencies, while rolling statistical features such as moving averages and rolling standard deviation are computed to identify volatility and trend patterns. Daily returns are calculated to understand percentage price change over time. Data normalization using MinMax scaling is performed to ensure that all features contribute proportionally during future modelling stages.

The project emphasizes building a structured and analysis-ready dataset that can later support machine learning forecasting techniques such as regression or deep learning-based models. By focusing on systematic preprocessing and transformation of financial time-series data, the study aims to improve model reliability and enhance interpretability in future predictive tasks.

3. BACKGROUND & PROBLEM STATEMENT

3.1 Background

Gold price prediction has been widely studied in financial analytics due to its direct relationship with global economic stability. Previous research highlights that factors such as inflation rates, currency exchange fluctuations, crude oil prices, and international political conditions influence gold prices. Statistical modelling and machine learning techniques are used by financial analysts to study gold price volatility and forecast market trends.

3.2 Problem Statement

Historical gold price data often contains non-stationary patterns, missing values, and noise, which restrict the effectiveness of machine learning forecasting models.

3.3 Need to Address the Problem

Conceptual Need: Gold price forecasting provides a structured framework to apply time-series analysis and machine learning techniques to understand temporal financial data. It enables the study of statistical properties such as trend, seasonality, and volatility while supporting the application of forecasting models such as ARIMA, SARIMA, Prophet, and deep learning architectures like LSTM. These techniques help capture non-stationary behaviour and improve predictive accuracy using real-world financial datasets.

Commercial Need: Reliable gold price predictions support investment planning, portfolio diversification, and risk management strategies for investors, traders, financial institutions, and

jewellery industries. Accurate forecasting assists businesses in optimizing pricing strategies, minimizing financial risk, and improving profitability through informed decision-making.

Social Need: Gold serves as a secure investment and savings instrument, especially during periods of economic instability, inflation, and geopolitical uncertainty. Forecasting gold price trends helps individuals make safer financial decisions, supports wealth preservation, and enhances financial literacy among investors and households.

Domain-Based Need: The problem demonstrates the practical implementation of data science and financial analytics methodologies. It highlights the importance of preprocessing, feature engineering, and time-series modelling in financial market prediction. The project also strengthens understanding of real-world applications of machine learning techniques in commodity price forecasting and economic trend analysis.

3.4 Relevance

The project supports investors, financial institutions, and economic analysts by providing a structured dataset for analyzing gold price trends. The preprocessing techniques implemented in this project help improve prediction accuracy and allow efficient financial decision-making. The study also demonstrates the importance of feature engineering in time-series machine learning applications.

4. LEARNING OBJECTIVES AND OUTCOMES

4.1 Objectives

- Analyse historical gold price data to identify temporal patterns and trends.
- Apply data preprocessing and feature engineering techniques to prepare time-series financial data.
- Implement a machine learning forecasting model to predict future gold prices.
- Evaluate model performance using statistical and classification-based metrics.
- Interpret forecasting results to derive meaningful financial insights.

4.2 Expected Outcomes

1. Identification of significant historical trends and temporal dependencies in gold price data
2. Development of a clean, structured, and feature-enhanced dataset suitable for time-series modelling.
3. Creation of a trained forecasting model capable of generating gold price predictions.
4. Performance assessment of the model using accuracy and directional prediction measures.

5. Analytical understanding of gold price behaviour to support financial decision-making.

5. DATASET DESCRIPTION

The dataset used in this project is a real-world public financial dataset containing historical gold futures price data collected from Yahoo Finance using the Python yfinance library. The dataset provides daily trading information of gold prices and is suitable for time-series forecasting and financial analysis.

The dataset covers the period from January 1, 2010 to December 31, 2024, consisting of approximately 3,700 trading-day observations. Each row represents a single trading day and records price fluctuations and trading activity. Among the available attributes, the Close price is selected as the target variable since it represents the final trading value of gold for each day and reflects overall market behavior.

The dataset is structured as a time-series dataset where the Date column is converted into datetime format and used as the chronological index. The dataset contains minor quality issues such as missing values during market holidays and volatility-related noise, which are addressed during preprocessing.

Data Source

Yahoo Finance (2024). Gold Futures (GC=F) Historical Data. Retrieved using Python yfinance library.

Dataset Feature Description

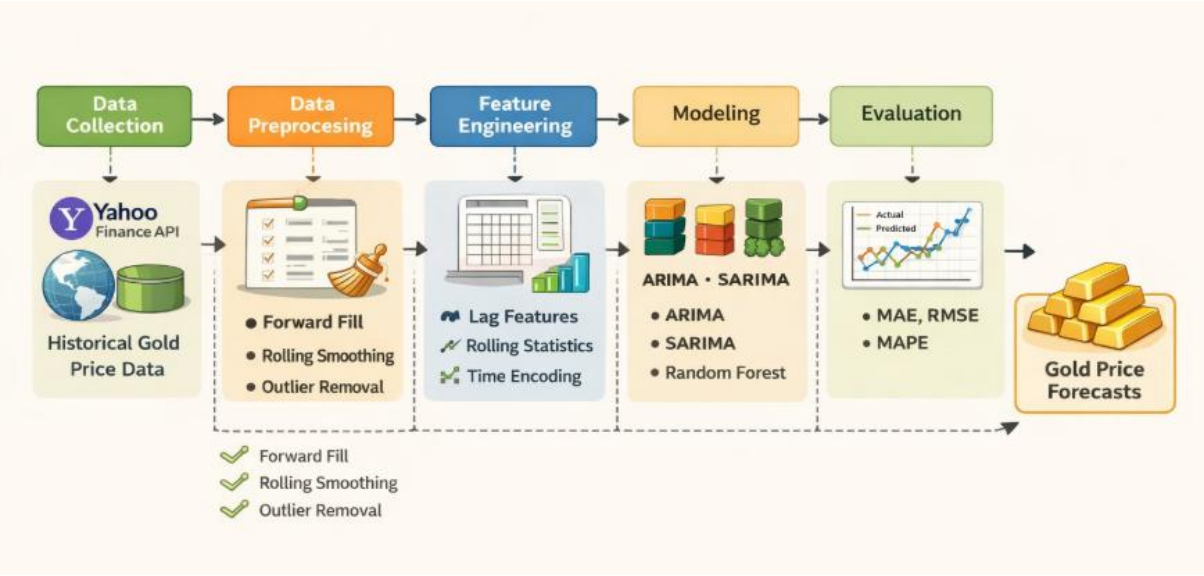
Sl. No	Feature Name	Description	Data Type	Issue Identified
1	Date	Trading day used as time index	datetime64	No missing values
2	Open	Opening gold price	float64	Missing values on non-trading days
3	High	Highest gold price of the day	float64	Occasional missing values
4	Low	Lowest gold price of the day	float64	Occasional missing values
5	Close	Final trading price (Target Variable)	float64	Minimal missing values
6	Volume	Number of contracts traded	int64/float64	Missing or zero values possible

Dataset Characteristics

- Number of records: ~3,700 observations
- Number of attributes: 6
- Data Constraints: Market volatility, noise, trend imbalance
- Data Quality Issues: Missing values handled using forward filling
- Datatype Summary (df.info()): Date → datetime64; Price attributes → float64; Volume → int64/float64

6. Methodology

This project follows the Knowledge Discovery in Databases (KDD) process to analyze and predict gold price movements using historical data. The KDD approach helps organize the work into clear steps, including data selection, preprocessing, transformation, modelling, and evaluation. Following this structured method helps ensure that the predictions are reliable and based on properly prepared data.



The analysis was carried out using Python and several data science libraries such as Pandas, NumPy, Statsmodels, Scikit-learn, Matplotlib, and Seaborn. These tools were used to clean, transform, visualize, and model the data. Since gold prices change over time, it was important to maintain the time sequence of the dataset while building prediction models. Therefore, the training and testing data were separated chronologically instead of randomly.

6.1 Data Selection

The dataset used in this project was collected using the Yahoo Finance API through the yfinance library. Gold futures price data was selected because gold is a widely traded commodity and

plays an important role in global financial markets. Predicting gold prices can help investors, analysts, and researchers understand market behaviour and make better financial decisions.

The dataset contains daily gold price data from the year 2010 to 2025. Using a long time period helps capture both short-term fluctuations and long-term trends in gold prices. Among the available attributes, the closing price was selected as the main variable for prediction. The closing price is commonly used in financial analysis because it represents the final market value of gold at the end of each trading day.

Other attributes such as opening price, highest price, lowest price, and trading volume were not included in the modelling process. These attributes were removed because the main goal of the project was to predict overall price trends rather than daily trading behaviour. To reduce short-term fluctuations and make trends easier to identify, a smoothed closing price was used as the final prediction target.

6.2 Data Preprocessing

Data preprocessing was necessary to improve the quality and reliability of the dataset before building prediction models.

Missing values were handled using forward filling. Since gold price data follows a time sequence, missing values were replaced using the most recent available price. This helps maintain continuity in the dataset.

To reduce noise in the data, a seven-day rolling average was applied. Gold prices often fluctuate daily due to market conditions, and smoothing helps highlight long-term trends instead of random daily changes.

Outliers were detected using the Interquartile Range (IQR) method. Extremely high or low price values were removed because they could affect model accuracy and lead to incorrect predictions.

Since the dataset contains continuous numerical values rather than categories, there were no class imbalance issues in the data. Pandas and NumPy were mainly used for preprocessing tasks, while visualization libraries were used to verify data quality.

6.3 Data Transformation

Data transformation techniques were used in the prediction models to understand patterns in the gold price data.

Lag features were created to include past price values as inputs. These included one-day, seven-day, and thirty-day lagged price values. These features allow models to learn how past price movements influence future prices.

Rolling statistical features such as rolling mean and rolling standard deviation were also created. The rolling mean helps represent local price trends, while rolling standard deviation helps capture volatility.

Time-related features like month and day of the week were extracted from the dataset index. These features help models identify seasonal patterns in gold price movements.

Since all features were numerical, encoding techniques were not required. The gold price values were scaled using MinMaxScaler to convert values between 0 and 1. Scaling helps machine learning models learn faster and perform better.

Only relevant engineered features were kept in the dataset to avoid unnecessary complexity and reduce the chances of overfitting.

6.4 Data Mining

The project uses regression-based time series forecasting techniques to predict gold prices. Three models were implemented to compare performance.

The ARIMA model was selected because it is commonly used for time series forecasting and can capture relationships between past values and prediction errors. The SARIMA model was used as an extension of ARIMA to include seasonal patterns that occur in gold price data. The Random Forest model was selected as a machine learning approach that can capture nonlinear relationships between features.

The ARIMA model works by analysing previous gold prices and identifying patterns that can be used for forecasting. SARIMA improves this by including seasonal components that help model repeating market trends. Random Forest builds multiple decision trees using engineered features such as lag variables and statistical measures, and the final prediction is generated by combining outputs from these trees.

The dataset was divided into training and testing sections based on time. Models were trained using historical data and tested using future price values to simulate real-world forecasting conditions.

6.5 Interpretation and Evaluation

Model performance was evaluated using three main metrics.

1. Mean Absolute Error (MAE) measures the average difference between predicted and actual values. Lower MAE values indicate better model accuracy.
2. Root Mean Square Error (RMSE) measures the overall prediction error and gives more importance to larger errors. Lower RMSE indicates better model performance.
3. Mean Absolute Percentage Error (MAPE) measures prediction error as a percentage of actual values.

The results showed that the SARIMA model produced the lowest MAE and RMSE values, indicating that it gave the most accurate predictions. The ARIMA model produced higher error values, which suggests that it was not able to capture seasonal price patterns effectively. The Random Forest model performed better than ARIMA but slightly worse than SARIMA.

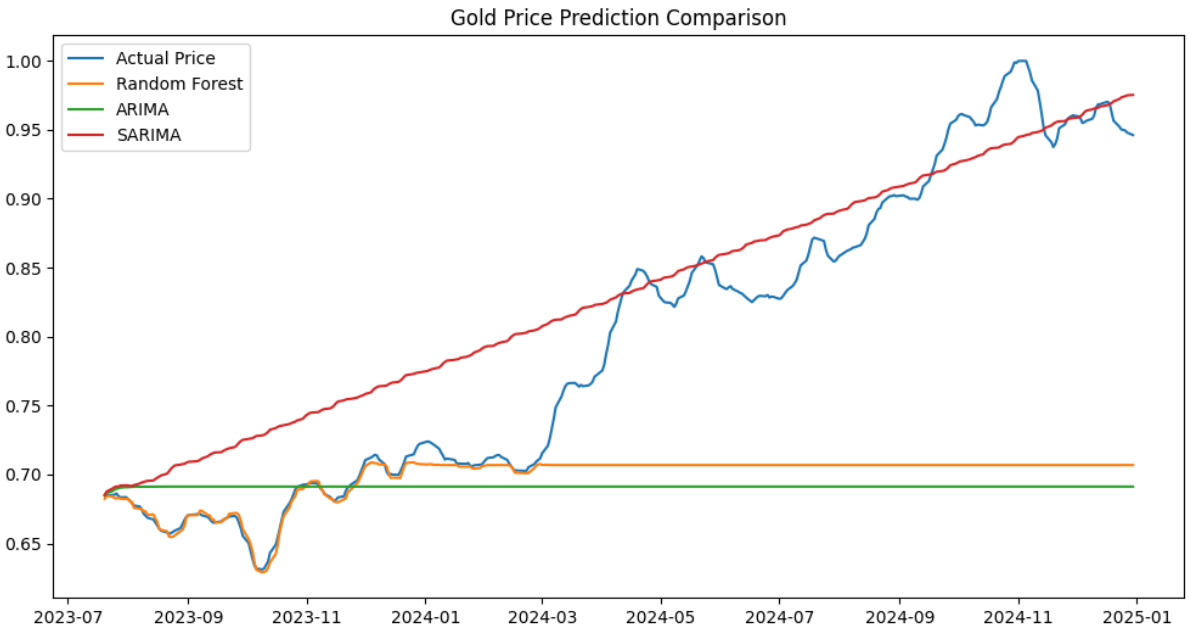
MAPE values were not available for ARIMA and SARIMA because scaling introduced zero values that affected percentage error calculations.

Prediction graphs showed that SARIMA predictions closely followed actual gold price trends, while ARIMA predictions showed larger deviations. Residual analysis for the Random Forest model showed that prediction errors were evenly distributed around zero, indicating stable predictions.

7. Results and Key Insights

7.1 Results

The comparison of models showed clear differences in prediction performance. SARIMA achieved the lowest error values and produced predictions that closely matched actual gold price data. Random Forest produced moderate results, while ARIMA showed the highest prediction error among the three models.



Visualization plots confirmed these findings by showing that SARIMA predictions aligned most closely with real price trends.

7.2 Key Insights

Seasonal behaviour plays an important role in gold price prediction, and models that capture seasonality perform better. Lag-based feature engineering helps models understand price

movement patterns. Machine learning models can identify complex relationships in data but may struggle to capture long-term seasonal behaviour. Data smoothing helps reduce noise and improves model stability.

8. Challenges, Pitfalls and Drawbacks

One of the major challenges in this project was the high volatility of financial data. Gold prices can change suddenly due to economic or political events, which makes prediction difficult. Removing outliers sometimes risks removing valid extreme market movements.

Scaling the dataset created difficulty while calculating percentage-based error metrics. Another limitation is that time series models rely heavily on historical trends and may not perform well if future market conditions change significantly.

Model accuracy also depends on the amount and quality of historical data available. Regular model retraining may be required to maintain performance over time.

9. Conclusion

This project successfully applied time series and machine learning models to predict gold price movements using historical data. The KDD process helped organize the project into structured steps, making the analysis systematic and reliable.

Among the tested models, SARIMA made the most accurate predictions because it captured both seasonal and trend patterns. Random Forest produced reasonable predictions using engineered features, while ARIMA showed comparatively lower accuracy.

The study shows that seasonal time series models are effective tools for financial price forecasting and highlights the importance of feature engineering in improving model performance.

10. Discussion Questions

1. How would including economic indicators such as inflation or currency exchange rates affect gold price prediction?
2. Could deep learning models improve forecasting accuracy compared to traditional time series models?
3. How would the model perform if real-time streaming data were used instead of historical data?
4. What would happen if hybrid models combining statistical and machine learning methods were used?

5. How can the model be adapted to handle sudden economic disruptions?

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APPENDIX – PROJECT METADATA

- **Difficulty Level:** Intermediate
- **Domain:** Finance and Financial Analytics
- **Dataset Type:** Real-time secondary dataset
- **Tools Used:** Python, Pandas, NumPy, Scikit-learn, Matplotlib, Statsmodels
- **KDD Focus:** Preprocessing, Feature Engineering, Modelling, Evaluation, Interpretation
- **Team Contribution:**
  - **Abdullah** handled data collection, modelling, and visualization.
  - **Vaishnavi** handled data preprocessing and documentation.
    Both members collaborated during discussions and result interpretation.